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PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	The Effects of Aggregate Culture in Modeled Microgravity and Partial Gravity on Endocrine Differentiation of Induced Pluripotent Stem Cells (iPSC)		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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	The long-term goal of this proposal is to study the effects of modeled microgravity and partial gravity on aggregate (organoid) cultures of induced pluripotent stem cells (iPSCs). Our working hypothesis, based on some preliminary data, is that hypogravity, i.e., modeled microgravity and/or partial gravity, will enhance iPSC differentiation into the endodermal lineages. Specifically, we will compare the efficiency of two distinct microgravity gravity / partial gravity simulating devices, i.e., the Rotating Wall Vessel (RWV) bioreactor and Random Positioning Machine (RPM) to modulate the maintenance of iPSCs stemness and the directed differentiation of iPSC into definitive endoderm. In testing our hypothesis, we developed the following specific aims: Specific Aim 1. To characterize the differentiation of iPSC into definitive endoderm in 2D monolayer culture in normal, and simulated partial- (Moon, Mars) and microgravity; and in 3D aggregate suspension culture in partial- (RPM) and microgravity (RPM and RWV). Specific Aim 2. Examine the underlying genetic, epigenetic, molecular, and biochemical mechanisms of stem cell differentiation and behavior that are affected by aggregate culture, hypoxia, and hypogravity; and identify specific molecular and signaling factors important in modulation of differentiation.
Task Description:	In these studies, we will use biochemical, immunological, and molecular biological approaches to analyze the differential expression of a set of lineage-specific specific marker genes and gene products under altered gravity conditions. Subsequently, we will identify specific genetic, epigenetic, and molecular mechanisms that underlie and are influenced by partial- and microgravity and compare these biologic factors with those from induced iPSC differentiation, in both aggregate and monolayer culture, in standard gravity. The latter data will be analyzed by systems- / bioinformatics-based analytical approaches, such as principle component analysis and network analysis, which will help to identify specific hypogravity-sensitive cellular signaling molecules and pathways that may be differentially affected by varying levels of partial/microgravity, and which could be further explored in actual space experiments. In addition to having available both hypogravity simulating devices, we have introduced several technical innovations, such as modifications in the RWV bioreactors, enhanced oxygenation capabilities in RPM culture, and a high-resolution optical monitoring system that allows real-time visualization of intracellular signaling events. We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
Task Progress:	1. Objectives: Short- and long-term habitation in hypogravity can induce cellular and systemic alterations that can be detrimental to the health of astronauts. Earth-based, modeled microgravity (MMG) systems using rotary bioreactors and the random positioning machine provides unique opportunities to evaluate these effects. We proposed to examine the behavior of pluripotent stem cells cultured as three-dimensional (3D) cell aggregates under MMG with and without directed differentiation to endoderm, a transitional stage between stem cells and cells of the respiratory and gastrointestinal tracts, as well as associated organs such as pancreas, liver, and thyroid. 2. Research Impact. Stem cell behavior and differentiation in MMG, and the specific advantages of three-dimensional (3D) cell culture (compared to traditional 2D monolayer culture), represent an important, under- examined field of inquiry. First, a better understanding of stem cell behavior and differentiation in microgravity should have considerable impact on NASA's Space Biology initiative, in terms of the adaptation to and potential amelioration of the detrimental effects of microgravity. Second, 3D culture systems have been shown to better emulate in vivo biology and behavior compared to traditional 2D monolayer culture, and the examination of these systems may contribute to enhancing technical and medical advancements on Earth. Finally, the development of human induced pluripotent stem cells (hiPSC) provides a relevant clinical platform to examine the effects of space travel on human physiology. 3. Research Progress. General Research Design: We proposed to culture hiPSC in MMG as well as altered gravity conditions akin to the gravity on Moon and Mars, and examine the following culture conditions: 1) 2D versus 3D aggregate culture, 2) normal gravity versus MMG, and 3) directed differentiation of pluripotent stem cells into definitive endoderm. Because of initial difficulties culturing hiPSC (see below, as also reported in last year's report), we concomitantly examined mouse embryonic stem cells (mESC). To that end, we generated embryoid bodies (EBs) and seeded them into the rotating wall vessel (RWV) bioreactor (BR). We then analyzed the cells and cell-aggregates using histology, flow cytometry, and RT-qPCR. We have made considerable progress with both of these cell lines, and several new findings have been realized and described at scientific conferences. mESC: Normal culture and directed differentiation in both 3D and microgravity culture (as described in last's year's report) demonstrated distinct differences between traditional static 2D culture and dynamic 3D culture in NASA's RWV bioreactors. These differences suggest the advantages of 3D culture and indicate specific effects of the dynamic MMG environment on stem cell behavior. To date, we have presented these results at two scientific conferences (see below). Furthermore, we have used the past 2 months of "shelter-in-place" and "working from home" to analyze our data and prepare the results for publication in a peer-reviewed journal. hiPSC: The primary pitfall and initial delay arose a mycoplasma contamination in the hiPSCs we received from an outside source. We had been working with the cells described in the grant application for more than 6 months before we were notified of the contamination and discounted all results. While we treated the hiPSC, we initiated a similar project with mESC as described above. Even after the cells were treated, we had difficulty culturing and passaging them. As an alternative, we purchased a new cell line from ATCC, and all the data presented here was generated from that cell line. A secondary pitfall was difficulty in culturing both cell lines, human and murine, in the RWV using single-cell approaches. To overcome this pitfall, we first modified our protocol to generate embryoid bodies (EBs) before seeding them in the bioreactor. Finally, after several attempts with complex protocols for directed definitive endoderm (DE) differentiation, based upon the results from mESC, we are now using a single factor-based approach with the small molecule activin A.

	Protocol. Based upon our results with mESC, we used the same protocol for hiPSC with a few notable exceptions. First, due to difficulties culturing the cells in the RWV bioreactor, we began using the random positioning machine (RPM; Airbus) for our MMG experiments. We cultured cells on the RPM in fluorinated ethylene propylene (FEP) bags in an in-house, 3D printed rack. Second, we added two additional culture conditions to evaluate 2D culture in microgravity in MM and DM. Third, we extended the differentiation protocol to 10 days and increased the activin concentration to 100 ng/ml. In the initial set of studies, we evaluated the MMG setting of the RPM, subsequent studies will address altered gravity conditions akin to reduced gravity on Mars and Moon. Initial results: In 2D culture, we saw similar morphology between cells in the FEP bags and those in tissue culture flasks, both in terms of normal cell colonies and those plated as single cells in preparation for directed DE differentiation. We also examined three pluripotency markers (nanog, oct3/4, and Sox2) and three endoderm markers (FoxA2, Sox17, and CXCR4), but, due to the sudden cessation of research activities in the wake of the COVID-19 pandemic, not all have markers been analyzed in all conditions. Conclusions and Future Approaches. The protocols had been optimized and data were being generated, when all lab functions were suspended in March 2020 due to COVID-19. While our data are preliminary, they suggest that microgravity alone directly influences hiPSC behavior and differentiation. Based on our results we will request a one year no-cost extension to continue and conclude our studies. According to the university's plans for returning to the labs, we expect resuming our research activities in a graded fashion by the middle-end of June. Our plan is to repeat the preliminary experiments with hiPSCs under MMG and altered gravity conditions and then perform more detailed genetic and proteomic analyses in order to identify mechanisms, as per our original proposal.
Bibliography Type:	Description: (Last Updated: 09/28/2023)
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